DIRECT SUMMANDS OF SUMS OF QUOTIENTS OF THE RING BY PRINCIPAL IDEALS, CHARACTERIZED BY A LIFTING PROPERTY FOR SHORT EXACT SEQUENCES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The hypothesis that the intersection of a submodule with the multiples of an element equals the multiples of the submodule is what makes the lifting argument for the quotients work. used at proof lines 37601-37607. Proves the first implication by an explicit lift constructed from the divisibility of the image of one.
- The canonical presentation of the module as the cokernel of the map from the direct sum over the pairs of an element and a map out of the corresponding quotient. used at proof lines 37610-37622. Produces the short exact sequence to which the second condition is applied in the converse.
- A relation in the canonical presentation can be corrected by an explicit class in a single summand, making the corrected tuple map to the original element. used at proof lines 37622-37634. Completes the converse by exhibiting the element whose image is the given one.

Lemma 0.2. *Let P be a module over a ring R. The following are equivalent*

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(1) *P is a direct summand of a direct sum of modules of the form R/fR, for $f \in R$ varying.*
(2) *for every short exact sequence $0 \to A \to B \to C \to 0$ of R-modules such that $fA = A \cap fB$ for all $f \in R$ the map $\mathrm{Hom}_R(P, B) \to \mathrm{Hom}_R(P, C)$ is surjective.*

Proof. Let $0 \to A \to B \to C \to 0$ be an exact sequence as in (2). To prove that (1) implies (2) it suffices to prove that $\mathrm{Hom}_R(R/fR, B) \to \mathrm{Hom}_R(R/fR, C)$ is surjective for every $f \in R$. Let $\psi : R/fR \to C$ be a map. Say $\psi(1)$ is the image of $b \in B$. Then $fb \in A$. Hence there exists an $a \in A$ such that $fa = fb$. Then $f(b - a) = 0$ hence we get a morphism $\varphi : R/fR \to B$ mapping 1 to $b - a$ which lifts ψ.

Conversely, assume that (2) holds. Let I be the set of pairs (f, φ) where $f \in R$ and $\varphi : R/fR \to P$. For $i \in I$ denote (f_i, φ_i) the corresponding pair. Consider the map

$$B = \bigoplus\nolimits_{i \in I} R/f_iR \longrightarrow P$$

which sends the element r in the summand R/f_iR to $\varphi_i(r)$ in P. Let $A = \mathrm{Ker}(B \to P)$. Then we see that (1) is true if the sequence

$$0 \to A \to B \to P \to 0$$

is an exact sequence as in (2). To see this suppose $f \in R$ and $a \in A$ maps to fb in B. Write $b = (r_i)_{i \in I}$ with almost all $r_i = 0$. Then we see that

$$f \sum \varphi_i(r_i) = 0$$

in P. Hence there is an $i_0 \in I$ such that $f_{i_0} = f$ and $\varphi_{i_0}(1) = \sum \varphi_i(r_i)$. Let $x_{i_0} \in R/f_{i_0}R$ be the class of 1. Then we see that

$$a' = (r_i)_{i \in I} - (0, \ldots, 0, x_{i_0}, 0, \ldots)$$

is an element of A and $fa' = a$ as desired. □